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Dae Yeon Kim Ph.D.

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EDUCATION

Ph.D.	**Seoul National University**, Seoul, Korea School of Chemical and Biological Engineering Thesis: "A study on particle deposition and clogging dynamics of colloidal suspensions in complex flow" (Advisor: Prof. Kyung Hyun Ahn)	Aug 2023
B.S.	**Sogang University**, Seoul, Korea Department of Chemical and Biomolecular Engineering Graduation honors, *Summa Cum Laude*	Feb 2017

PROFESSIONAL EXPERIENCES

Assistant Professor	Department of Chemical and Biomolecular Engineering Chonnam National University, Yeosu-si, Jeollanam-do, Korea	Mar 2026 – Present
Postdoctoral Scholar	Department of Chemical Engineering Stanford University, Stanford, CA, USA (Advisor: Prof. Sho Takatori)	Feb 2025 – Jan 2026
Postdoctoral Scholar	Department of Chemical Engineering University of California, Santa Barbara, Santa Barbara, CA, USA (Advisor: Prof. Sho Takatori)	Mar 2024 – Jan 2025
Postdoctoral Researcher	Center for Coating Materials and Processing Seoul National University, Seoul, Korea (Advisor: Prof. Kyung Hyun Ahn)	Sep 2023 – Jan 2024

RESEARCH EXPERIENCES

Research Skills

Microfluidics	• Design and fabrication of microchannel • Fluid visualization and PIV • Pressure drop measurement
Rheometry	• Rotational rheometer (ARES, AR-G2, DHR) • Extensional rheometer (Capillary Breakup Extensional Rheometer) • Passive/Active microrheology (Optical tweezer)
Capillary breakup	• Measurement of elongational property of complex fluids (polymer solution, colloidal suspension) • Direct visualization of thinning filament surface via high-speed camera
Optical tweezer	• Measurement of inter-particle interaction (e.g. Hydrodynamic interaction) • Microrheology of soft matter
Chemical labeling	• Soft vesicle labeling (SUV, GUV) • Colloids and bacteria labeling
Bacterial culturing	• Characterization and labeling • Quantitative imaging analysis

Analysis tool
- Image processing and quantitative analysis (MATLAB, Python)
- CFD simulation (COMSOL)

Research Interests

Complex materials and soft matter physics
- Colloidal suspension rheology
- Fluid-mediated interactions between colloids
- Active matter rheology and control
- Colloidal fouling onto a membrane surface (complex interactions between foulants and membrane)
- Flow instability of viscoelastic fluid

Heterogeneity control of industrial slurry (e.g. lithium-ion batteries, fuel cells, CMP slurry)
- Stability control of colloidal slurry
- Characterization of slurry for quality control

Data-driven machine learning classification
- Characterization of complex fluid
- Clogging and fouling prediction

Image-based rheological properties measurement
- Discovery of constitutive relations for complex fluids
- Dynamics of soft materials with active forcing

PUBLICATIONS

Peer-reviewed Journal Articles

* equal contribution † co-corresponding

13. Haegyo Jeong, Minsun Hwang, **Dae Yeon Kim**†, Seon Yeop Jung†, Influence of fluid viscosity and cavity length on particle transport and deposition in expansion-contraction microchannels, *Korea-Australia Rheology Journal*, **2026**, 38, 335–354. [Link]

12. **Dae Yeon Kim***, Sachit G. Nagella*, Saksham Malik, Nayeon Park, Jaewook Nam, Eric S.G. Shaqfeh, Sho C. Takatori, Colloidal hydrodynamic interactions in viscoelastic fluids, *Soft Matter*, **2025**, 21, 8035–8048. [Link]

11. Minsun Hwang, **Dae Yeon Kim**, Kyung Hyun Ahn, Sung Sik Lee, Seon Yeop Jung, Particle deposition in colloid-polymer suspensions: Complex interplay between flow characteristics and polymer interactions, *Physics of Fluids*, **2025**, 37, 073389. [Link]

10. **Dae Yeon Kim***, Sachit G. Nagella*, Kyu Hwan Choi, Sho C. Takatori, Direct experimental measurement of many-body hydrodynamic interactions with optical tweezers, *Physical Review Fluids*, **2025**, 10, 064301. [Link]

9. Woobin Yi*, **Dae Yeon Kim***, Howon Jin, Sangwoong Yoon, Kyung Hyun Ahn, Early detection of pore clogging in microfluidic systems with 3D convolutional neural network, *Separation and Purification Technology*, **2025**, 359, 130428. [Link]

8. Young Jin Lee, **Dae Yeon Kim**, Kyung Hyun Ahn, Particulate fouling simulation in unit micropore using a hydrodynamically coupled Lagrangian framework, *Desalination*, **2025**, 593, 118255. [Link]

7. Minsun Hwang, **Dae Yeon Kim**, Seon Yeop Jung, Kyung Hyun Ahn, Effect of fluid viscosity on the deposition of colloidal particles in an expansion-contraction microchannel flow, *Nihon Reoroji Gakkaishi*, **2024**, 52 (4), 231–240. [Link]

6. Wonsik Shin, Joung Sook Hong, **Dae Yeon Kim**, Si Yoon Kim, Kyu Hyun, Jun Dong Park, Kyung Hyun Ahn, Effect of cellulose nanocrystals on the emulsion stability and rheological properties of microalgal Pickering emulsions, *Algal Research*, **2024**, 83, 103731. [Link]

5. Young Jin Lee, Howon Jin, **Dae Yeon Kim**, Seunghoon Kang, Kyung Hyun Ahn, Lagrangian-based simulation method using constrained Stokesian dynamics for particulate flows in microchannel, *Journal of Fluid Mechanics*, **2024**, 990, A3. [Link]

4. **Dae Yeon Kim**, Seon Yeop Jung, Young Jin Lee, Howon Jin, Kyung Hyun Ahn, From successive deposition to clogging of poly(styrene) particles in the cross-flow through a T-shaped microchannel, *Separation and Purification Technology*, **2023**, 322, 124263. [Link]

3. **Dae Yeon Kim**, Seon Yeop Jung, Young Jin Lee, Kyung Hyun Ahn, Effect of colloidal interactions and hydrodynamic stress on particle deposition in a single micropore, *Langmuir*, **2022**, 38 (19), 6013–6022. [Link]

2. **Dae Yeon Kim**, Kyung Hyun Ahn, Effect of operating conditions on the sessile drop oscillation of PEO solution in the capillary thinning process, *Nihon Reoroji Gakkaishi*, **2021**, 49 (2), 127–133. [Link]

1. Youngseok Kim, **Dae Yeon Kim**, Joung Sook Hong, Kyung Hyun Ahn, Agglomerate breakup of destabilized polystyrene particles under a cross-channel planar extensional flow, *Langmuir*, **2018**, 34 (38), 11454–11463. [Link]

Submitted/in-progress

3. **Dae Yeon Kim**, Free-vibration mechanism and viscoelastic damping of sessile-drop oscillation in capillary thinning, *submitted.*

2. **Dae Yeon Kim**, Colloidal hydrodynamic interactions: from bulk rheometry to rotation-based optical tweezer measurements, *accepted.*

1. **Dae Yeon Kim**, Izzie Allydice, Sho C. Takatori, Colloidal hydrodynamic interaction in bacterial active bath, *in preparation.*

PRESENTATIONS

International Conference

9. Fluid-mediated rotational coupling and hydrodynamic interactions in colloidal particles using optical trapping, *APS Global Physics Summit (APS March Meeting 2025)*. 16 – 21 Mar 2025, Anaheim, CA, USA.

8. Successive deposition of poly(styrene) particles in cross-flow through T-shaped microchannel, *The 19th International Congress on Rheology (ICR 2023)*. 29 July – 4 Aug 2023, Athens, Greece.

7. Effect of salt concentration and flow rate on clogging dynamics at the single micro-pore. *The 8th Pacific Rim Conference on Rheology (PRCR 2023)*. 15 – 19 May 2023, Vancouver, BC, Canada.

6. Effect of salt concentration and flow rate on clogging dynamics at the single micro-pore. *The 93rd Annual Meeting of the Society of Rheology (SoR 93rd Annual Meeting)*. 9 – 13 Oct 2022, Chicago, IL, USA.

5. Effect of salt concentration and flow rate on clogging dynamics at the single micro-pore. *11th Korea-Australia Rheology Conference (KARC 2021)*. 24 – 25 Nov 2021, Virtual.

4. The effect of operating condition and rheological properties of PEO solution on the sessile drop oscillation during elasto-capillary thinning. *International Union of Materials Research Societies-International Conference in Asia (IUMRS-ICA 2020)*. 23 – 26 Feb 2021, Virtual.

3. Effect of rheological properties of PEO solution on the sessile drop oscillation during the early stages of elasto-capillary thinning transition. *The 15th International Workshop for East Asian Young Rheologists (IWEAYR 2020)*. 8 – 10 Jan 2020, Changchun, China.

2. Effect of rheological properties of PEO solution on the sessile drop oscillation during the early stages of elasto-capillary thinning transition. *Annual European Rheology Conference (AERC 2019)*. 8 – 11 Apr 2019, Portoroz, Slovenia.

1. Agglomerate breakup of destabilized polystyrene particles under a cross-channel planar extensional flow. *The 14th International Workshop for East Asian Young Rheologists (IWEAYR 2019)*. 23 – 26 Jan 2019, Nagoya, Japan.

Others (2023 – Present)

5. Particle Deposition, Pore Clogging, and AI-Assisted Prediction of Membrane Fouling. *KSIEC Special Symposium: AX in the Carbon-Neutral Era, Korean Society of Industrial and Engineering Chemistry*. 10 – 11 Sep 2026, Yeosu, Korea

4. Probing hydrodynamic interactions and rheology in complex fluids using optical laser tweezers. *2026 Annual Meeting of the Korean Society of Rheology*. 7 – 8 May 2026, Seoul, Korea

3. Prediction of membrane fouling via particle transport phenomena and ai-based analysis. *2026 Spring Conference of the Korean Society for Energy*. 15 – 17 Apr 2026, Yeosu, Korea

2. Direct experimental measurement of hydrodynamic interactions using optical tweezers. *Shaqfeh Group Seminar, Stanford University*. May 2025

1. A study on particle deposition and clogging dynamics of colloidal suspensions in complex flow. *EEPL Group Seminar, Dankook University*. Aug 2023

TEACHING AND MENTORING

Teaching (S:Spring, F:Fall)	Transport Phenomena (Chonnam National University)	2026F
	Chemical Reaction Engineering II (Chonnam National University)	2026F
	General Chemistry I (Chonnam National University)	2026S
	Chemical Reaction Engineering (Chonnam National University)	2026S
	Creative Academy: Basic Chemical Engineering (Janghoon High school, Seoul)	Mar 2018 – Jan 2019
TA	Transport Phenomena (Seoul National University)	Spring. 2018
	Introduction to Chemical Engineering (Seoul National University)	Fall. 2019
Mentoring	2025 ChemE REU Program Mentee: Izzie Allydice (Stanford '28)	Jul 2025 – Present
	EEPL Group, Dankook University Mentee: Minsun Hwang (B.S.–M.S. Integrated Program)	Mar 2022 – Jul 2025

AWARDS AND HONORS

Best Paper Award	DONGJIN Best Paper Award (supported by DONGJIN SEMICHEM CO., LTD.)	Jul 2022

Best Presentation Award	Best Presentation Award of 15th IWEAYR (The 15th International Workshop for East Asian Young Rheologists)	Jan 2020
Academic Scholarship	Scholarship for academic excellence (Sogang University)	Sep 2013 – Aug 2016

OTHER ACTIVITIES

Member	Young Engineers Honor Society (YEHS)	Jan 2014 – Present
Student Leader	Department of Chemical and Biomolecular Engineering Sogang University, Korea	Jan 2014 – Mar 2014

REFERENCES

Dr. Kyung Hyun Ahn	Professor, School of Chemical and Biological Engineering, Seoul National University, Seoul, Korea	ahnnet@snu.ac.kr
Dr. Sho Takatori	Associate Professor, Department of Chemical Engineering, Stanford University, Stanford, CA	stakatori@stanford.edu
Dr. Seon Yeop Jung	Assistant Professor, Department of Chemical Engineering, Hongik University, Seoul, Korea	syjung@hongik.ac.kr